

STEMsters Robotics Competition (SRC)

The word "SUMMIT" is rendered in a large, bold, sans-serif font. The letters "S", "U", "M", and "I" are a vibrant teal color, while the letters "M", "I", and "T" are a lighter, sky-blue color. The letters are closely spaced and have a slight shadow effect.

2026 Game Manual

October 24, 2026
Arcadia, CA

Version 1.0 | September 21, 2026
STEMsters is a student-led 501(c)(3) nonprofit

Summit

- **The game.** Two autonomous robots on a small raised hill. Push the other robot off, or be higher up when time runs out.
- **The robot.** 10 x 10 cm at the start, 500 g or less, no height limit, controlled by a micro:bit. Everything else is up to the team.
- **The match.** Three minutes. The referee's field controller starts and stops both robots by radio.
- **The team.** One to four students.
- **The day.** Qualification matches, then a single-elimination bracket. Schedule and awards are on the SRC website.

Climb together. Respect your opponents, the referees, and the volunteers.

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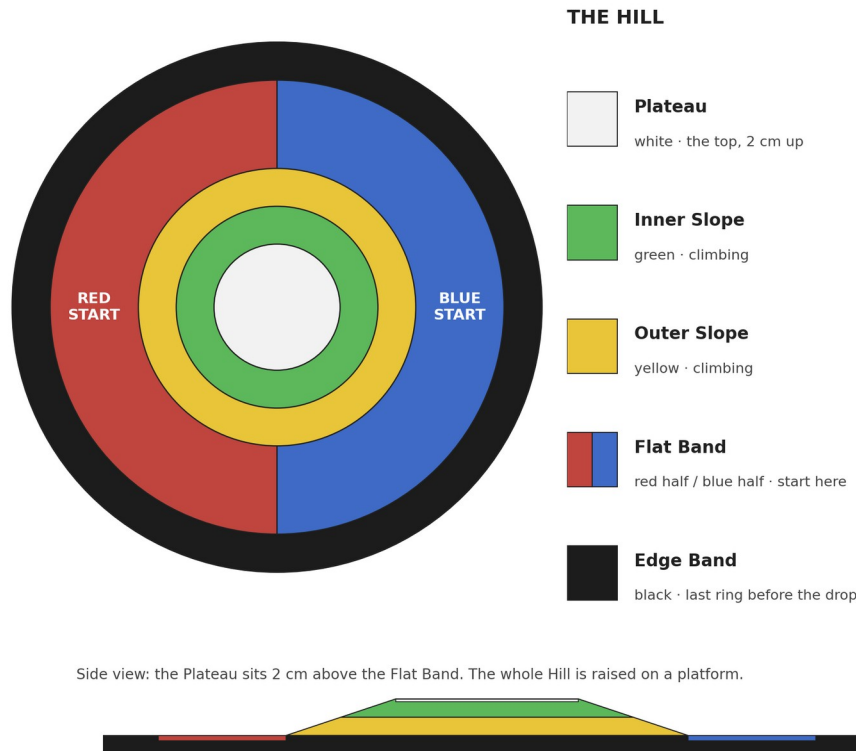
1. About SRC and Summit

The STEMsters Robotics Competition (SRC) is an autonomous robotics competition for middle and high school students in Southern California, run by STEMsters, a student-led 501(c)(3) nonprofit. Summit is the 2026 game. This is Version 1.0, for the October 24, 2026 event.

Summit is autonomous sumo on the Hill: push the other robot off, or be higher up when time runs out. Once a match starts, no one touches or controls the robots.

Registration and kit pricing: robotics-comp.vercel.app.

2. The Hill



A 3D-printed PLA mound, 42 cm across, on a raised platform. The outer edge is a drop: a robot that drives off cannot get back on. Each ring is a different color. From the top:

- **Plateau (white).** The flat top, 2 cm above the base.
- **Inner Slope (green) and Outer Slope (yellow).** The two sloped rings.
- **Flat Band (red half, blue half).** The flat base. Robots start here.
- **Edge Band (black).** The outermost ring. Its outer edge is the boundary.

Dimensions are in Appendix A.

3. The match

3.1 Starting the match

1. **Red places first.** Colors come from the schedule: random in qualification; in the bracket the lower seed is red. Place anywhere on your half, facing any direction. The robot may touch only your color of the Flat Band and the Edge Band beside it.
2. **Start.** The field controller counts down five seconds and sends START to both robots.
3. **Backup.** If the controller fails, both handlers press their start buttons on the referee's count.

3.2 End of the match

- **Ring-out.** Any part of a robot touches anything other than the Hill or the other robot. The referee stops both robots; the other robot wins. Both out with no clear first: replay.
- **Time.** At 3:00 the controller stops both robots and the result is judged at that moment. Higher up wins: the robot whose drive wheels or treads touch the ring closest to the center. Only drive wheels or treads count. Same ring: tie.
- **Dead stall.** Neither robot changes position for 30 seconds; spinning in place or pushing without moving counts as not changing. The referee counts the last ten seconds aloud, ends the match, and judges it as at time.

4. Scoring

- **Result.** Win, loss, or tie, per 3.2.
- **Tie.** Both drivetrains in the same ring at time or dead stall.
- **Qualification.** Win 3, tie 1, loss 0. A forfeit is a loss and a win for the opponent. Rank by points, then wins. If exactly two teams remain tied and played each other, that winner ranks higher; otherwise, random draw.

5. Robot rules

R1 Size. At the start, the robot must fit inside the official size gauge, a box lowered over it from above. The gauge is 10.0 x 10.0 cm plus a small clearance and is the final test. No height limit.

R2 Expansion. No intentionally detaching, dropping, or launching parts. A part that falls off by accident does not end the match and is not a ring-out.

R3 Weight. 500.0 g or less in any configuration, battery included.

R4 Controller. One micro:bit running the team's code makes the robot's strategic and movement decisions. No other processor may make those decisions, and no second micro:bit.

R5 Radio. No remote control. The only wireless allowed is receiving START and STOP from the field controller through the official code. Robots never transmit. Bluetooth off. No sensors, beacons, or markers off the robot.

R5a Official code. Published for MakeCode at robotics-comp.vercel.app. Every robot runs it, unmodified except for the team number. It holds the robot still until a START addressed to its number, stops the motors on STOP, and provides the manual start with its five-second delay. Team code runs inside it.

R6 Manual start. A physical start button with a five-second delay, used only when the controller fails.

R7 Power cutoff. A switch or connector, no tools needed, that cuts drive power.

R8 Batteries. Commercially made, unmodified, secured, undamaged, used within spec. No voltage limit.

R9 Team number. Emailed at registration. Write it legibly on the robot and enter it once in the official code.

R10 Entanglement. No nets, string, ribbon, or flexible parts meant to snare.

R11 Wheels. No adhesives, treated tires, liquids, powders, or loose material. A wheel fails inspection if paper pressed onto it lifts with the robot.

R12 Magnets. None, except inside motors and sensors.

R13 Weapons. No spinners, hammers, saws, blades, spikes, or edges sharp enough to cut skin.

R14 Sensor interference. Nothing that emits light, sound, or radio to jam, blind, or spoof the other robot's sensors.

R15 Also banned. Flame, heat, chemicals, projectiles, glass.

R16 One robot. No second robot. Spare parts only.

R17 Changes. Reinspect after any change that could affect size, weight, legality, or safety. Code changes need no reinspection.

R18 Match color. Every robot must be able to show red or blue for its match, visible from above: a flag, a light, a swappable marker, anything switchable between matches. A robot without one gets an SRC clip from the referee before the match. The clip does not count toward size or weight.

6. Team rules

E1 Team. One to four students, grades 6 to 12.

E2 Roster. Registration names are the roster, shown on the SRC website as first name and last initial. Only rostered students handle the robot at the field and at inspection, one handler per match.

E3 Level. Middle school: grades 6 to 8. High school: 9 to 12. Any high school student makes it a high school team.

E4 Adult. Each team registers a coach or supervising adult, 18 or older, who is present for the whole event, signs the waiver and student-built pledge at check-in, and is responsible for the team. No adult, no matches.

E5 Forms. Every student needs a consent and release form signed by a parent or guardian. Submit it online by October 17 or bring the signed form to check-in. No form, no matches.

E6 Student-built. Students must be the primary designers, builders, and programmers, and must be able to explain their robot.

E7 Registration. robotics-comp.vercel.app/register. Fee, deadline, and kit ordering are there. Team numbers are emailed at registration.

7. Tournament

T1 Inspection. Pass before your first match (Appendix C). Students make the fixes. Not passed by the deadline: forfeit until passed. Referees may reinspect any robot at any time.

T2 Format. Middle and high school teams compete together: one ranking, one bracket.

T3 Qualification. The same number of matches for every team, random opponents. The number is set on the day.

T4 Bracket. Every inspected, checked-in team enters. Single elimination seeded by rank, byes to the top seeds. Each pairing is best of three; a tied match is replayed.

T5 Be on time. Queue at your field before the call. One minute late is a forfeit.

T6 Calls. A rostered student may question a call at the field right after the match. The head referee's decision is final. Adults do not approach referees.

T7 Exhibition. Exhibition matches do not count toward anything.

T8 Awards. Champion: the bracket winner. Middle School Champion: the middle school team that went furthest in the bracket, ties by qualification rank; if a middle school team is Champion, it goes to the next-best middle school team. Finalists and semifinalists are recognized.

Schedule, calibration time, and the exhibition field: robotics-comp.vercel.app.

8. Safety

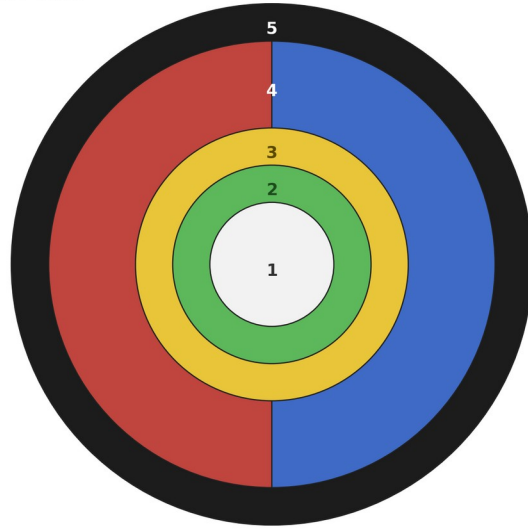
- Charge batteries only in the charging area and stay with them. Report damaged or hot batteries at once.
- Safety glasses when using tools. No soldering irons, power tools, flame, sparks, or fumes.
- In an emergency, follow event staff.

9. Violations

- **Not ready.** Late to the field, not inspected, or an illegal or unsafe robot: forfeit until fixed. An illegal item found after a match makes that match a loss.
- **Cheating.** Remote control, signals to a robot, intentional damage, a robot that is not student-built: match loss, award ineligibility, or disqualification, at the head referee's discretion.
- **Conduct.** Warning, then loss of the next match, then disqualification. Applies to adults too.
- **Interference.** A person touches a robot or the field mid-match: replay, or a loss for that person's team. A field failure or referee error that changes the result: replay.

Appendix A. Field specification

TOP VIEW

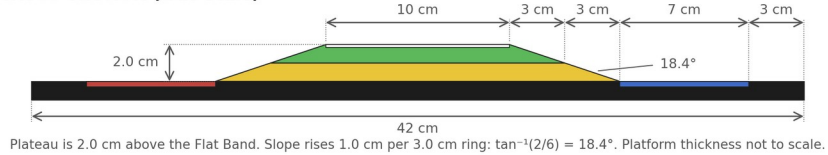


ZONES (outer diameter)

-  **Zone 1 Plateau**
10 cm dia · flat · +2.0 cm
-  **Zone 2 Inner Slope**
16 cm dia · 18.4°
-  **Zone 3 Outer Slope**
22 cm dia · 18.4°
-  **Zone 4 Flat Band**
36 cm dia · flat · start zones
-  **Zone 5 Edge Band**
42 cm dia · flat · boundary

42 cm overall

CROSS-SECTION (true scale)

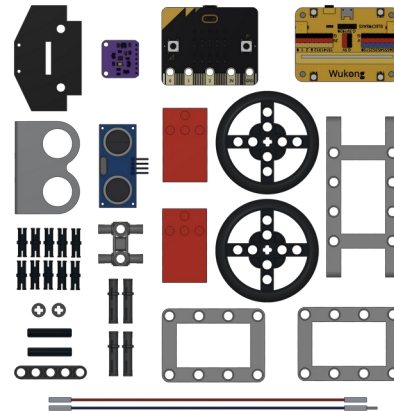
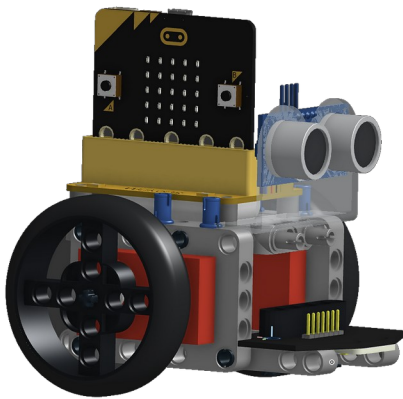


Ring	Inner radius (cm)	Outer radius (cm)	Surface	Color
Plateau	0	5.0	Flat, 2.0 cm above base	White
Inner Slope	5.0	8.0	18.4 degree slope, rises 1.0 cm	Green
Outer Slope	8.0	11.0	18.4 degree slope, rises 1.0 cm	Yellow
Flat Band	11.0	18.0	Flat, base level	Red half, blue half
Edge Band	18.0	21.0	Flat, base level	Black

- 3D-printed PLA, raised on a platform.
- A practice Hill comes with the kit and is sold separately. Print files, filament per ring, and reference readings from the kit color sensor: robotics-comp.vercel.app. Readings vary with sensor, height, and lighting; calibrate on a real field.

Appendix B. Official kit

Optional. Everything for the official robot except the BBC micro:bit, which is required and sold separately. Ships unassembled; teams build it from the build guide and program it in MakeCode.



- ELECFREAKS Wukong expansion board with built-in battery
- Two Geeservo 2 kg 360-degree motors on M1 and M2
- HC-SR04 ultrasonic sensor
- TCS34725 color sensor
- LEGO Technic chassis parts and wheels
- Practice Hill, same filament as the competition fields
- Starter code with the official start/stop code (R5a)

Ordering and pricing: robotics-comp.vercel.app.

Appendix C. Inspection checklist

1. Fits the official size gauge (R1).
2. 500.0 g or less, heaviest configuration (R3).
3. One micro:bit makes the robot's decisions (R4).
4. Team number written on the robot (R9). Inspection sticker applied on passing.
5. No entanglement devices, adhesives, magnets, liquids, loose material, weapons, sharp edges, glass, or sensor-jamming devices (R10 to R15). Paper test on every wheel (R11).
6. Nothing designed to detach (R2).
7. Shows red and blue, or has a spot for the SRC clip (R18).
8. Battery commercially made, unmodified, secured, undamaged (R8). Wiring secured, no exposed conductors.
9. Power cutoff works (R7).
10. Manual start button and five-second delay work (R6).
11. Official code with the correct team number: stays still until a test START, stops on STOP (R5, R5a, R9). Team confirms no transmitting and Bluetooth off.